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Simulation and Prediction of Ship Trajectories

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Abstract

Understanding and predicting maritime vessel movement is critical for ensuring navigational safety, efficient port operations, and effective environmental oversight. This project presents a simulation framework for ship trajectory modeling, combining physics-based dynamics with insights drawn from historical Automatic Identification System (AIS) data. By integrating real-world navigation patterns with environmental variables, the model aims to replicate realistic ship behavior under varying conditions. Through this hybrid approach, we bridge theoretical modeling with practical data-driven insights to contribute meaningfully to maritime intelligence and safety. The evaluation demonstrates that the model can reproduce historical vessel tracks with high fidelity, achieving path similarity scores up to 0.96 in confined inland waterways and above 0.92 in open-water cases when full environmental forcing is included. Cross-track error was typically within 10–20 m for best cases, with worst-case deviations of 40–50 m reflecting limitations in environmental input resolution rather than deficiencies of the physics model.

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1 Introduction

Maritime navigation is a complex domain involving numerous dynamic factors such as vessel behavior, sea conditions, and traffic density. With the global increase in marine traffic, there is a growing need to understand and predict ship movement accurately. This project focuses on simulating and forecasting ship trajectories by leveraging historical AIS (Automatic Identification System) data and physics-based modeling approaches.

This topic lies at the intersection of transportation analytics, maritime safety, and data-driven simulation. It is relevant in the context of smart ports, autonomous navigation systems, and real-time maritime surveillance. Integration of real-world vessel data with simulation models supports more informed decisions in both civilian and commercial maritime operations.

Working in this field holds significant practical importance. Accurate ship trajectory prediction can:

- Improve port traffic management by optimizing docking and scheduling.
- Contribute to environmental protection by identifying risky navigation near sensitive zones.
- Enhance maritime safety by detecting anomalies and preventing collisions.
- Support the development of autonomous maritime systems, which rely heavily on reliable trajectory forecasting.

As the maritime industry pushes toward automation and digitization, advancements in trajectory modeling will play a critical role in ensuring safe, efficient, and sustainable marine operations.

2 Task Description and Project Goals

The primary objective of this project is to develop, implement, and evaluate a physics-based model for the accurate prediction of ship trajectories. The core problem this project addresses is the significant influence of environmental factors such as wind, water currents, and visibility on vessel movement, which is often underestimated or ignored in conventional trajectory prediction models. By failing to account for these external forces, existing models can produce overly optimistic and inaccurate forecasts, particularly in geographically complex or congested waterways. This work aims to bridge this gap by creating a simulation framework that integrates real-world vessel kinematics from Automatic Identification System (AIS) data with corresponding meteorological and hydrological data.

The ultimate goal is to produce a validated model along with an Interactive UI capable of simulating future vessel paths with high accuracy, thereby enhancing maritime safety, traffic

management, and the development of autonomous navigation systems. To achieve this, the project is structured into several phases:

- **Exploratory Data Analysis (EDA):** The initial phase involves the comprehensive analysis of historical AIS data from the two designated study areas: the inland Rheinhafen region and the tidal waters near Cuxhaven. This step includes data cleaning, filtering, and the statistical examination of key kinematic parameters like Speed Over Ground (SOG), Course Over Ground (COG), and heading. The objective is to identify baseline vessel behavior, traffic patterns, and characteristic maneuvers specific to each geographical context.
- **Weather and Environmental Analysis:** This stage focuses on fusing the processed AIS data with concurrent environmental data, including wind speed and direction, water current velocity, and visibility conditions. The goal is to quantitatively assess the impact of these factors on vessel dynamics. This analysis will seek to establish clear correlations, such as the effect of cross-winds on vessel drift (leeway) and the influence of head- or tail-currents on SOG, validating patterns identified in previous research.
- **Physics-Based Model Development:** Leveraging the insights from the preceding analyses, this core phase involves the development and implementation of a physics-based ship motion model (e.g., a Manoeuvring Modeling Group (MMG) model). The model will be designed to explicitly incorporate the quantified environmental forces as inputs. The objective is to create a dynamic simulation engine that can predict a vessel's trajectory over time based on its initial state (position, speed, heading) and the prevailing environmental conditions.
- **Model Evaluation and Validation:** In this step, the predictive accuracy of the physics-based model will be rigorously assessed. Simulated trajectories generated by the model will be systematically compared against the ground-truth trajectories recorded in the historical AIS data. The evaluation will utilize appropriate error metrics to quantify model performance and will specifically analyze its effectiveness under varying degrees of environmental hindrance (e.g., calm vs. stormy conditions).
- **UI Development and Refinement:** The final stage of the project is the creation of an intuitive User Interface (UI) for interacting with the simulation model. This UI will serve as a visualization and analysis tool, allowing users to input starting and ending coordinates and simulate the resulting trajectory prediction on a map.

3 Scientific/Technical Status at Project Start

3.1 Literature Review

Recent literature highlights the broad utility of Automatic Identification System (AIS) data for maritime navigation analysis, encompassing traffic pattern detection, encounter scenario

classification, risk modeling, environmental impact quantification, and machine-learning-driven behavioral assessment. Researchers have used AIS to evaluate vessel interactions such as overtaking, crossing, and head-on encounters, to study the influence of weather and current, and to connect these factors to performance metrics including speed reduction, trajectory deviation, and collision risk [4, 7, 13, 17, 21].

AIS-based behavioral analyses have produced several key findings regarding environmental effects. Zhou et al. [24] analyzed the Rotterdam straight fairway and showed that currents aligned with or against the head of the ship strongly influence the speed over ground, while crosswinds mainly induce leeway, measured as the angle between the head and the course over the ground. Their later work, along with Zhao et al. [25], introduced hydrodynamic current models and statistical thresholds to define unhindered and hindered conditions. Shu et al. [18, 19] contributed detailed analyses of vessel speed, course, and path deviations in port environments, highlighting how external conditions and traffic encounters shape behavior. Together, these studies established robust empirical approaches for disentangling environmental impacts from navigational interactions.

However, most prior AIS-based work has focused on single environments, especially Rotterdam, leaving open the question of how well findings generalize across different geographical contexts. Comparative studies are less common, though Xiao et al. [21] demonstrated the value of regional traffic comparisons, and probabilistic collision risk modeling based on AIS has been advanced by Goerlandt and Kujala [7], Montewka et al. [13], and Park et al. [17]. Against this background, our study applies encounter filtering, visibility thresholds, and size-based stratification to two new contexts: the confined inland Rhine and the open tidal waterway of Cuxhaven. Linking these empirical analyses to a physics-based MMG model provides a contextualized application of prior methods rather than an entirely novel methodology, while also addressing the underexplored role of geography in mediating environmental hindrance in vessel trajectories.

Prior studies such as Zhou et al. [24, 25] also identified directions for future research. Most analyses were limited to nearly straight waterways, excluding the influence of geometry. They explicitly noted that port areas or waterways with more complex layouts should be studied to assess the effects of geometry and terminal approaches. They further observed that sailing direction (inbound vs. outbound) and distance to destination can influence speed choice, but remained underexplored. Finally, they suggested a comprehensive nautical traffic model that integrates wind, current, and vessel characteristics, offering predictive value for researchers and practical utility for port authorities.

Alongside extending these statistical methods, a prominent alternative approach involves purely data-driven trajectory prediction using machine learning and deep learning [12, 22]. These techniques treat vessel movement as a complex time series problem, learning navigational patterns [11] directly from historical AIS data without relying on explicit hydrodynamic equations. Models such as Long Short-Term Memory (LSTM) [20] networks have been widely applied due to

their ability to capture temporal dependencies in sequential data. More recently, Transformer-based architectures have gained traction for their superior capacity to model long-range dependencies and complex interactions between vessels and their environment, making them powerful tools for predicting future positions over extended horizons.

While a thorough review of these data-driven techniques was conducted, their implementation was deemed unsuitable for this project due to two primary constraints. First, deep learning models are notoriously data-hungry, requiring vast and diverse datasets to train effectively and avoid overfitting. Sourcing a sufficiently large and high-quality AIS dataset, complete with corresponding high-resolution weather data for the two specific regions of interest, posed a significant challenge. Second, the process of developing, training, and fine-tuning these complex models is computationally expensive and time-intensive, falling outside the allocated timeline for the project. The risk of insufficient data leading to a poorly generalized model, combined with the time constraints, made this approach impractical.

To complement AIS-based statistical studies, physics-based maneuvering models provide a mechanistic way to simulate ship behavior under environmental influences. A well-established framework is the MMG (Maneuvering Mathematical Group) model proposed by Ogawa and Kasai [15], which decomposes vessel motion into contributions from the hull, propeller, and rudder. This modular structure has made MMG widely adopted for maneuvering simulations. Propeller thrust is typically expressed through the thrust coefficient K_T as a function of advance ratio J, derived from propeller series such as the Wageningen B-series [2], while rudder forces and control laws are formulated in Fossen's framework for marine craft hydrodynamics and control [5].

Environmental effects are also incorporated into physics-based methods. Wind loads are often estimated using the empirical Fujiwara model [6], while the International Towing Tank Conference (ITTC) provides recommended procedures for current load estimation [8]. Line-of-Sight (LOS) guidance has been emphasized as a robust path-following method, with Lekkas and Fossen [9] showing stable convergence properties under disturbances. In parallel, AIS-based studies such as Shu et al. [19] and Zhou et al. [25] have used LOS centerlines as baselines for quantifying deviations in vessel motion, underscoring the bridge between empirical and model-based approaches.

The LOS method offers a geometric reference against which vessel position can be projected, enabling calculation of the cross-track error (CTE), the primary signal for trajectory correction. By generating heading references directly from LOS geometry rather than fixed waypoints or time-parameterized paths, LOS ensures smooth and stable convergence. Recent work has extended this foundation by applying optimization methods such as CMA-ES to fine-tune MMG model parameters against real-scale AIS trajectories [23].

Together, these physics-based approaches complement AIS statistical analyses by enabling controlled simulation of environmental forces and trajectory dynamics.

Our project extends prior research by analyzing two contrasting environments beyond the nearly straight ramp of Rotterdam: the confined inland Rhine and the open tidal estuary of Cuxhaven. This allows explicit evaluation of geographical and layout effects on vessel behavior, moving closer to the "more complex port areas" suggested in earlier work. By integrating AIS-weather empirical analysis with physics-based MMG modeling, the project evaluates ship responses to wind and current both statistically and mechanistically. In doing so, it advances the understanding of how environmental factors and geography jointly influence vessel trajectories, reinforcing the central role of weather and natural forces in maritime navigation analysis.

3.2 Historical AIS Dataset

For the **Cuxhaven region**, the dataset comprised **19,207,524 AIS messages** spanning 15 days of vessel activity. The main navigation-related message types (1, 2, 3, and 18) accounted for **18,336,783 records**, complemented by ship information messages (type 5: **741,033 records**; type 24: **128,814 records**). Within the main polygonal region of interest, **2,116,836 records** were retained, corresponding to **1,122 unique MMSI**. In a smaller sub-polygon for environmental impact analysis, **330,395 records** corresponding to **855 unique MMSI** were identified.

For the **Rhein Haven region**, a 7-day AIS dataset containing **339,600 records** and **274 unique MMSI** was available. Within the sub-polygon for environmental analysis, **26,123 records** corresponding to **219 unique MMSI** were extracted.

This raw dataset provided the empirical foundation for validating the simulation model across two distinct maritime domains.

Table 1: Summary of AIS datasets used for model validation.

Region	Duration	Total Records	Unique MMSI
Cuxhaven (main polygon)	15 days	2,116,836	1,122
Cuxhaven (sub-polygon)	15 days	330,395	855
Rhein Haven (main polygon)	7 days	339,600	274
Rhein Haven (sub-polygon)	7 days	26,123	219

3.3 Expertise of Team Members

Vineet: Python, Machine Learning, Research, Databases, Frontend Development, and Scientific Writing.
Nishanth: Python, Modeling, EDA, Frontend Development, and Research.
Aishwarya EDA, Databases, Data Analysis, Python, Research, and Preparing Presentations.

3.4 Tools and Libraries

Programming Language: Python(NumPy, Pandas, Matplotlib, Seaborn), SQL.
Modeling Libraries: SciPy (CubicSpline), Shapely, Geopy, Overpy.
Development and Database Tools: MySQL 8 with Workbench, Visual Studio Code (IDE), GitHub, Overleaf.
User Interface Development: Streamlit Community CLoud, Folium, Plotty.

4 Project Plan

This section outlines the strategic plan for the project, detailing the methodology, work packages, timeline, and allocation of responsibilities. The project is structured to ensure a systematic progression from foundational research to model development, validation, and final dissemination. The approach is iterative, allowing for refinement based on findings at each stage.

4.1 Work Packages, Task Descriptions and Assigned Responsibilities

The project is divided into seven distinct milestones, each with specific objectives and deliverables.The detailed project schedule is visualized in the Gantt chart below.

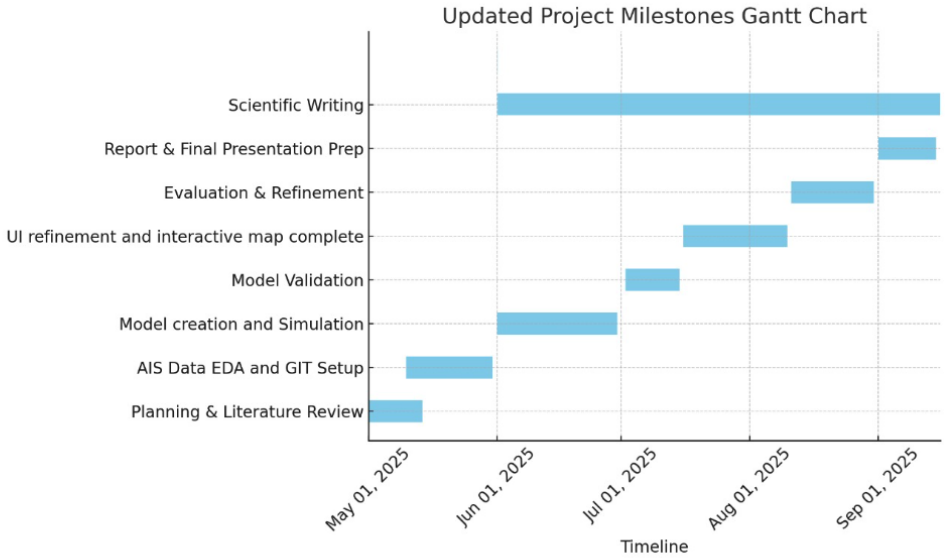


Figure 1: Estimated Project Timeline

M1: Planning & Literature Review Duration: 2 Weeks (Early May 2025)
Description: This initial phase focuses on establishing a solid theoretical foundation

and a clear project roadmap. Key activities include a comprehensive review of scientific literature on ship trajectory prediction, physics-based maneuvering models (e.g., MMG), the analysis of environmental impacts on navigation using AIS data, and data-driven approaches. The intended outcome is a refined project scope and a detailed understanding of state-of-the-art methodologies.

Assigned Responsibilities: All three members were tasked to review literature related to the project

M2: AIS Data EDA and GIT Setup Duration: 3 Weeks (Mid-May – Late May 2025)

Description: This milestone involves the acquisition and preparation of the core datasets. Historical AIS data for the Rhine and Cuxhaven, along with corresponding meteorological will be collected. An Exploratory Data Analysis (EDA) will be performed to identify data quality issues, understand traffic patterns, and derive key statistics. Analysis of Weather influence will be done to see environmental impacts in ship trajectories. Concurrently, a Github repository will be established to manage code and documentation. **Assigned Responsibilities:** All three members were tasked to do the initial EDA. Aishwarya was responsible for setting up the Github repository, data cleaning and analysis of environmental factors.

M3: Model Creation and Simulation Duration: 5 Weeks (Early June – Early July 2025)

Description: The central task of this work package is the development of the physics-based ship motion model. Based on the literature review, a suitable model (e.g., MMG) will be implemented. The core of this work involves integrating external environmental forces (wind and current) into the model's equations of motion to simulate their effect on the vessel's trajectory. An initial simulation environment will be built to run the model.

Assigned Responsibilities: Nishanth was responsible for the development of the model. Aishwarya developed the line-of-sight plan for the model.

M4: Model Validation Duration: 2 Weeks (Early July – Mid-July 2025)

Description: The objective of this phase is to rigorously assess the accuracy of the simulation model. Trajectories generated by the model under historical weather conditions will be quantitatively compared against the ground-truth vessel paths from the AIS data. Standard error metrics will be employed to measure performance, and the model parameters will be calibrated to minimize discrepancies.

Assigned Responsibilities: Nishanth was responsible for the development of the model validation. Vineet collaborated to validate edge cases for different starting and ending points.

M5: UI Refinement and Interactive Map Duration: 4 Weeks (Mid-July – Mid-August 2025)

Description: To facilitate the analysis and demonstration of the model, an interactive User Interface (UI) will be developed. This UI will feature a map-based simulation to display an animated simulation of the predicted trajectory of the physics-based model developed. It will allow the user to configure simulation parameters, such as starting

point, ending point and speed, to explore different scenarios.
Assigned Responsibilities: Vineet was responsible for this task.

M6: Evaluation & Refinement Duration: 2 Weeks (Mid-August – Late August 2025)
Description: This phase involves a final, holistic evaluation of the entire system (model and UI). The model's performance will be tested across a wide range of conditions to identify its strengths and limitations based on historical AIS data. Based on these findings, final refinements will be made to both the simulation engine and the user interface.
Assigned Responsibilities: Aishwarya and Nishanth collaborated to evaluate the physics-based model with the ground truth and refined the model, while Vineet was making refinements to the UI based on changes being made to the model.

M7: Scientific Writing, Reporting & Presentation Duration: 13 Weeks (Parallel, June – September 2025)
Description: This is an ongoing task that runs in parallel with the technical work packages. All methodologies, experiments, and results will be continuously documented. The final weeks will be dedicated to compiling the comprehensive project report and preparing the final presentation and demonstration of the project's outcomes.
Assigned Responsibilities: All three members are responsible to document their respective tasks for the project. Vineet is responsible for proofreading and maintaining the overall structure and tone of the report.

5 Project Execution and Scientific/Technical Results

5.1 Exploratory Data Analysis of AIS Data

Exploratory Data Analysis (EDA) was performed collaboratively by **all three team members** for the Rheinhafen and Cuxhaven regions. The primary dataset consisted of *Automatic Identification System (AIS)* records, which provide spatio-temporal information on vessel movements. The raw AIS data were first subjected to a cleaning process that included: removal of missing or duplicated entries, filtering of invalid coordinates outside the bounding boxes of the study regions, and conversion of timestamps into a consistent format suitable for temporal analysis. Noise introduced by static or incomplete records was also mitigated by applying filtering thresholds on vessel speed and position changes.

Following preprocessing, the data were examined using `NumPy` and `Pandas` for statistical summaries (e.g., vessel counts, distributions of vessel types, speed ranges, and message types). `Matplotlib` was employed to generate temporal and spatial plots, such as distributions of vessel density over time and histograms of speed over ground (SOG). `Folium` was used to construct interactive geospatial visualizations of ship trajectories, allowing for inspection of navigation patterns of both regions. This inspection could be done for a particular MMSI or for all MMSI IDs.

The EDA revealed distinct traffic characteristics between the two study sites. In the **Cuxhaven region**, the data showed a higher concentration of long-haul traffic, with dense inbound and outbound trajectories aligned with major shipping lanes. In contrast, the **Rheinhafen region** displayed more localized movement patterns, with shorter and slower trajectories consistent with port operations and inland navigation. These insights were critical in informing subsequent simulation model development and validation. They also guided the selection of bounding boxes used to constrain our modeling and simulation.

5.2 AIS Data Analysis under Wind and Current Conditions

To quantify the environmental influence on vessel behavior, we conducted a weather data analysis for both Cuxhaven (open-water approaches) and Rhein Haven (confined port-like waterways). AIS observations were aligned with wind and current data obtained from the Open-Meteo Historical Forecast API (Open-Meteo, 2025), and each vessel record was classified into hindered or unhindered states based on relative wind/current direction and magnitude thresholds adapted from Zhou et al. (2017). This enabled a structured comparison of vessel speed over ground (SOG) distributions under different environmental conditions. By applying the same methodology across both regions, we could contrast how open-water geometry and confined harbor layouts modulate the impact of wind and current, providing insight into size-dependent and directional behavioral patterns.

5.2.1 Data Cleaning

The raw AIS data were first subjected to a cleaning stage to ensure the reliability of subsequent analyses. Invalid or incomplete identifiers, such as corrupted MMSI/IMO numbers, were removed. Position reports falling over land were discarded as spurious, and records with invalid or missing timestamps were excluded. For temporal consistency across datasets, Unix timestamps were converted into UTC, with a specific calibration for the Cuxhaven region to match meteorological data alignment. This step follows best practices in AIS preprocessing to minimize noise and guarantee valid spatial-temporal positioning of vessels [25].

5.2.2 Encounter Filtering

Because our focus was on environmental influences rather than navigational interactions, we filtered out ship encounters. Using trajectory geometry and relative velocity constraints, we excluded AIS points corresponding to head-on, crossing, and overtaking scenarios. This ensured that changes in speed over ground (SOG) could be attributed primarily to environmental forcing rather than collision-avoidance maneuvers. A similar assumption was applied by [24], who restricted their analysis to non-encounter, good-visibility conditions.

5.2.3 Weather Data Alignment

To quantify environmental influences, we aligned AIS records with wind, current, and visibility data from the Open-Meteo Historical Forecast API [16]. Meteorological and hydrological variables were interpolated to match the AIS timestamp resolution. This approach provides time-synchronized forcing conditions for every vessel observation. In Zhao's empirical work, local hydrodynamic models and measurement stations were used to pair AIS with environmental inputs. By applying the Open-Meteo API, we achieved a comparable integration of AIS and environment, enabling condition-specific behavioral analysis [24].

5.2.4 Ship Size Categorization

Ships were categorized by beam width, as beam is a primary determinant of both aerodynamic (windage) and hydrodynamic (current) exposure. Following [24], four categories were defined as detailed in Table 2:

Table 2: Vessel Size Classification Based on Beam Width

Vessel Class	**Beam Width (m)**	**Examples / Description**
Small Craft	Beam < 10	Tugs, workboats
Small–Mid Cargo	$10 \leq$ Beam < 23	General small to mid-sized cargo vessels
Mid-Size Cargo	$23 \leq$ Beam < 33	Panamax-type vessels
Large Vessels	Beam ≥ 33	Wide-beam tankers and bulk carriers

This stratification allowed us to examine how different vessel classes respond to environmental loads while maintaining comparability with prior research.

5.2.5 Labeling Environmental Hindrance States

Each AIS observation was labeled by hindrance conditions from relative wind/current angles and intensity thresholds adapted from [24] as detailed in Table 3:

Table 3: Definition of Unhindered and Hindered Environmental Conditions

Factor	**Condition**	**Wind Type**	**Relative Angle**	**Speed**
Wind	Unhindered	Tailwind	$315° - 45°$	< 6 m/s
	Hindered	Headwind	$135° - 225°$	≥ 6 m/s
Current	Unhindered	Following	$\leq 45°$ or $\geq 315°$	< 0.37 m/s
	Hindered	Opposing	$135° - 225°$	≥ 0.37 m/s

This classification enabled systematic comparison of vessel behavior under unhindered vs hindered conditions and quantification of the environmental penalty in terms of SOG changes.

5.2.6 Study Site Selection and Polygon Definition

To define the spatial boundaries for analysis, we used the Overpass Turbo API (https://overpass-turbo.eu/) to query navigational and infrastructure features in both the Cuxhaven and Rhein Haven areas. The queries included categories such as:

- Navigational aids (buoys, beacons, lights)
- Fairways and waterways (`"waterway"="fairway"`)
- Infrastructure (docks, piers, harbors, marinas)
- Seamark objects (anchorages, mooring areas, ferry lines)

Using these data layers, we investigated the local geography and selected polygons that captured suitable segments of the waterways. The polygons were drawn to:

- Avoid high-traffic areas such as ferry lines, anchorage zones, and port basins,
- Exclude maneuvering-intensive regions with jetties, turning circles, or harbor basins,
- Include relatively straight fairway sections where vessel paths are dominated by environmental forces rather than complex maneuvering.

Additionally, fairway lines used for line-of-sight analysis were also fetched in the same manner, enabling us to better represent navigational constraints within the polygons.

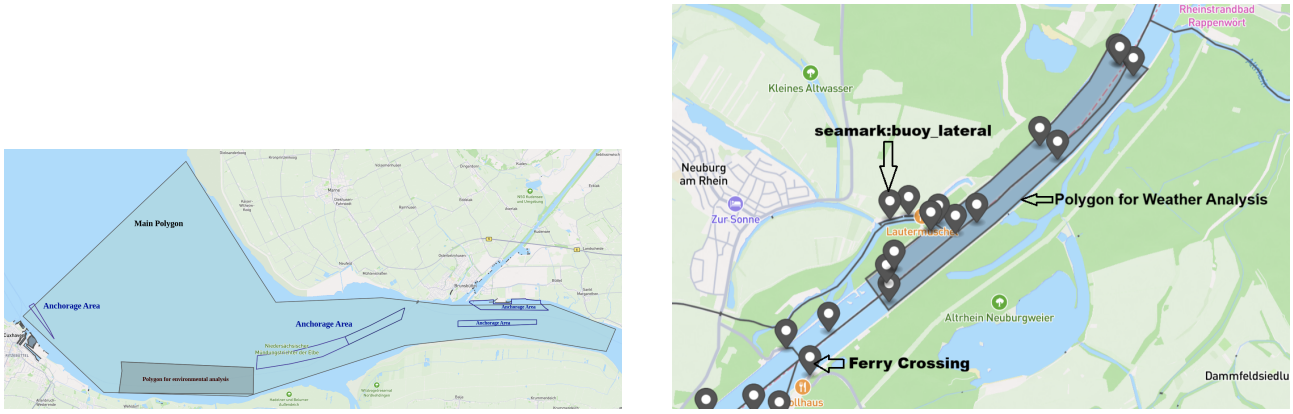


Figure 2: Study site polygons selected for simulation analysis: Cuxhaven estuary region (left) and Rhein Haven inland port (right).

This choice follows the methodology of [?], who explicitly analyzed a straight waterway to remove the confounding influences of channel curvature or port geometry.

5.2.7 Weather Data Analysis

Once the AIS dataset was prepared and hindrance states labeled, we performed the core weather analysis to evaluate the effect of wind and current on vessel speed over ground (SOG) and path behavior in both study regions. Median and percentile statistics were calculated for each condition, stratified by beam category and direction of travel (inbound vs outbound), following the analytical approach described in Zhou et al. (2017).

5.3 Findings of Weather Data Analysis

5.3.1 Cuxhaven Region

Cuxhaven represents an open-water approach where ships travel between the North Sea and the Elbe River entrance toward the Kiel Canal (Brunsbüttel). For the analysis, the vessel direction was categorized as **Inbound:** *Heading upriver toward Brunsbüttel / Kiel Canal.* **Outbound:** *heading seaward toward Cuxhaven and out to the North Sea.*

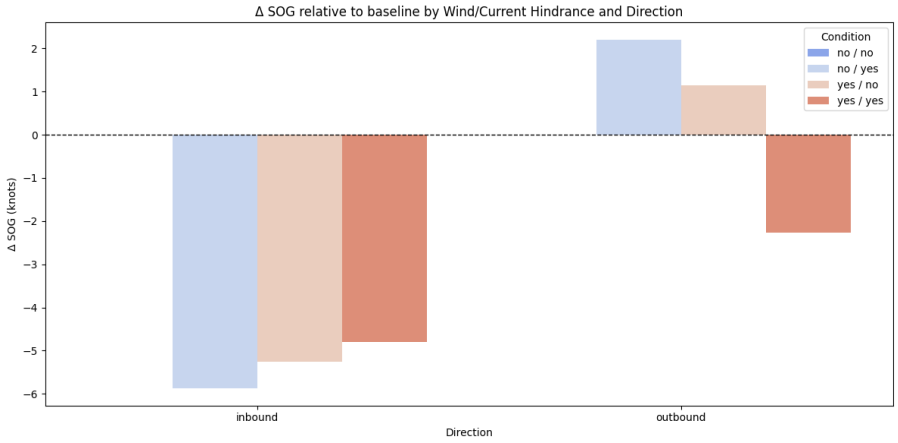


Figure 3: Cuxhaven: ΔSOG grouped by combined wind + current hindrance conditions.

Speed behaviour under hindrance: Figure 3 illustrates the effect of wind and current hindrance on vessel speed. It also shows the baseline wind effect: inbound vessels are slowed by 5–6 kn under headwinds, while outbound vessels maintain higher speeds but still suffer when both wind and current oppose them. This summarizes the **interaction of wind and current**, with ΔSOG grouped by combined condition (e.g. wind=yes / current=no). This highlights that simultaneous headwind + head current yields the strongest slowdown.

Beam category effects: Figures 4 and 5 show how beam size mediates the effect of hindrance.

Narrow-beam vessels (10–23 m) experience the strongest slowdowns under adverse conditions, especially with opposing currents (−2.6 kn). Moreover, *wider-beam vessels (23–33 m)* also show

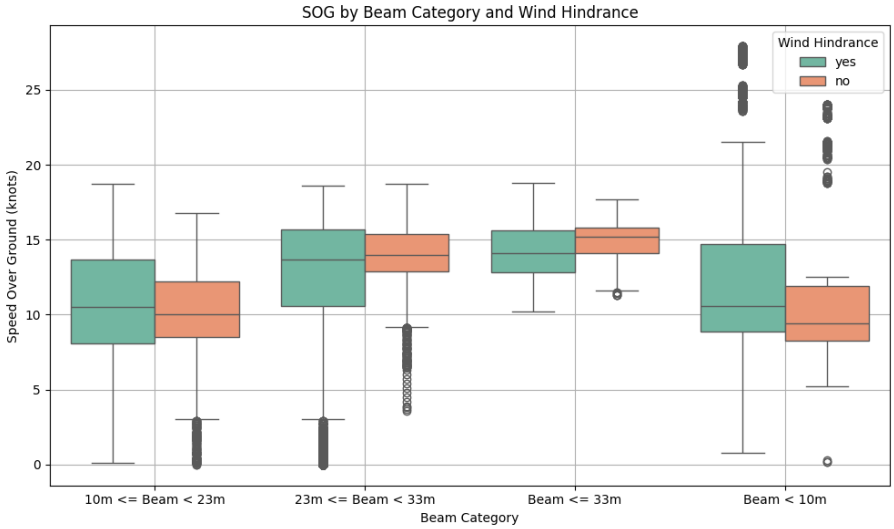


Figure 4: Cuxhaven: SOG by beam category under wind hindrance.

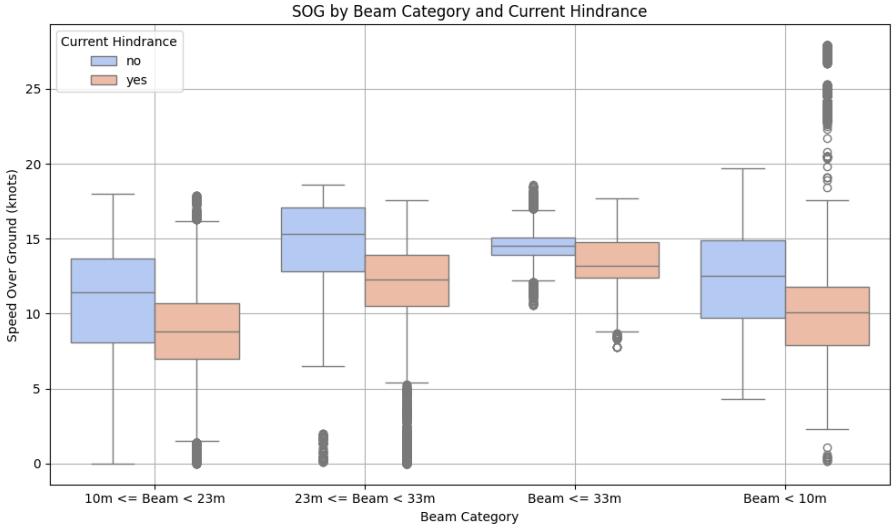


Figure 5: Cuxhaven: SOG by beam category under current hindrance.

significant slowdowns (∼2.5 kn) but are somewhat more resilient to wind forcing. Finally, *very small vessels (< 10 m)* are highly sensitive to both wind and current, reflecting lower inertia and stability.

Turn probability under current alignment: Figure 6 shows turn probability distributions by current alignment and vessel direction. Inbound ships consistently exhibit higher turn rates, especially under cross-current conditions (0.40%), followed by head currents. Outbound vessels turn less frequently, but still show sensitivity under cross-current flows.

Summary: The Cuxhaven analysis confirms that **currents dominate vessel behaviour**, while wind acts as an additional resistance factor. Inbound vessels toward Brunsbüttel are

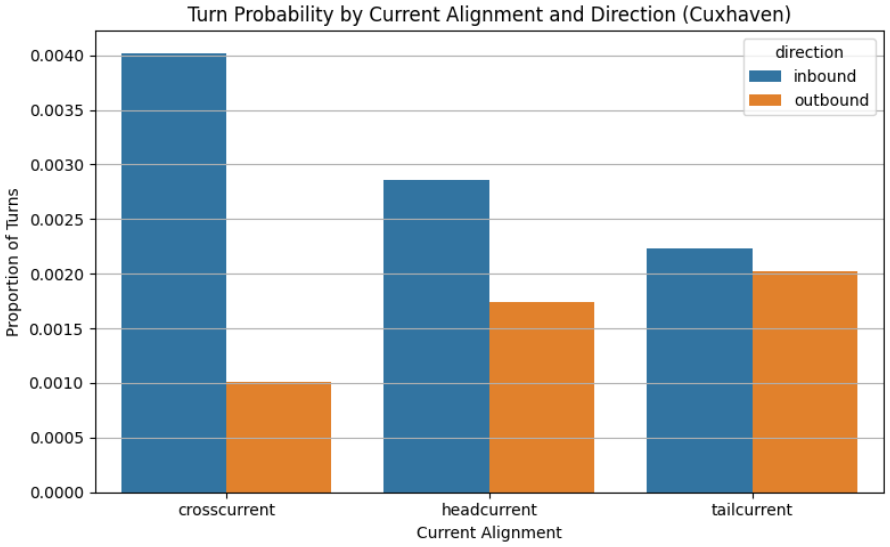


Figure 6: Cuxhaven: Turn probability by current alignment (inbound vs outbound).

particularly vulnerable, with speed reductions of 5–6 kn under head currents and elevated turn probabilities. Outbound vessels are less sensitive but still suffer marked slowdowns when headwinds and head currents coincide.

The regression analysis reinforces these findings by showing that:

- Tailwinds reduce the need for course corrections, leading to smoother trajectories.
- Head currents increase steering demands, especially for inbound vessels.
- Wider-beam ships are more exposed to environmental forcing, while high-speed vessels limit heading changes to preserve stability.

Comparison with Zhou's findings. Table 4 summarizes how our results align with, and extend, earlier work.

Aspect	Our Findings	Zhou's Findings
Wind effect on speed	Very small changes	Very small, inconsistent
Wind size sensitivity	Small vessels slightly faster; large ships $\sim$1 kn slower	Small vessels variable; large ships slightly slower
Current magnitude	Clear slowdown	Clear SOG impact
Current size pattern	Mid $>$ small $>$ large	Small $>\downarrow$ mid $>\uparrow$ large

Table 4: Comparison of our findings with Zhou's observations

Overall, both studies confirm that currents are the dominant driver of speed and maneuvering behavior in tidal approaches. Our analysis additionally highlights the disproportionate effect of currents on medium-beam vessels in Cuxhaven, extending Zhou's size-dependent patterns.

5.3.2 Rheinhafen Region

The Rheinhafen represents a confined inland waterway section, where current data were not available, and analysis focused on wind hindrance alone. Vessel direction was categorized into two categories. **Inbound**, *which defines heading upriver toward the port terminal* and **Outbound**, *heading downriver, leaving the port area.*

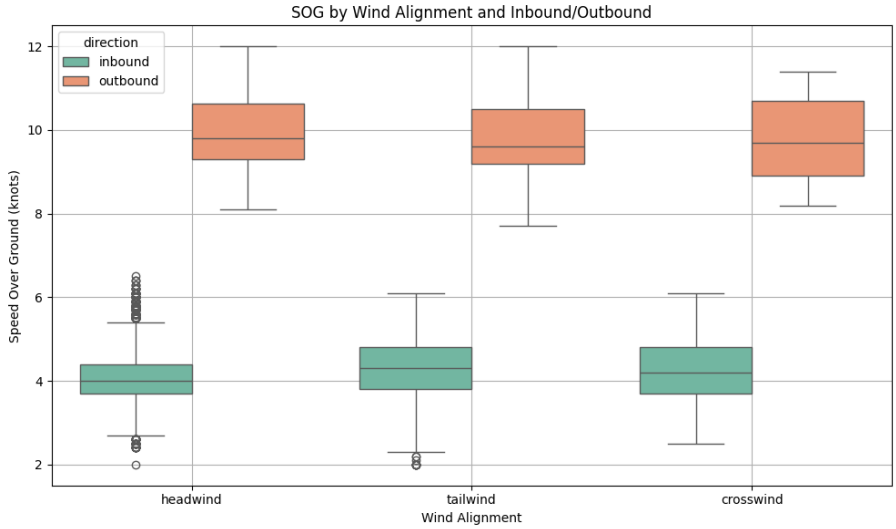


Figure 7: Rhein Haven: SOG by wind alignment and direction (inbound vs outbound).

Speed by wind alignment : Boxplots of SOG by wind alignment and direction (Figure 7) show that Inbound ships are consistently slow (∼4–5 kn), with headwinds producing the strongest slowdowns (−0.9 kn, 15–20% reduction). Outbound ships are much faster (∼10 kn) and show little to no reduction under wind hindrance (a slight +0.1 kn increase observed). Lastly, fleet-wide wind hindrance reduces median speeds by 4–6 kn (50–65%) compared to unhindered conditions. This confirms that *wind* is the dominant environmental constraint in this inland section.

Beam category effects: Boxplots of beam vs. wind hindrance (Figure 8) reveal strong size dependence. Small to mid-beam vessels (10–33 m) slow dramatically from 9–10 kn (unhindered) to 3–4 kn under headwinds. Very small vessels (< 10 m) are also highly sensitive, with speeds only around 4–5 kn under wind. Finally, larger vessels, though less represented, show vulnerability but less dramatic slowdowns.

Turn probability by wind alignment: Figure 9 shows turn probability stratified by wind alignment. Inbound vessels show the highest turn rates under cross-winds (5.5%), followed by tail-winds (3.8%) and head-winds (2.3%). However, outbound vessels show much lower turn rates overall (1–2% across categories).

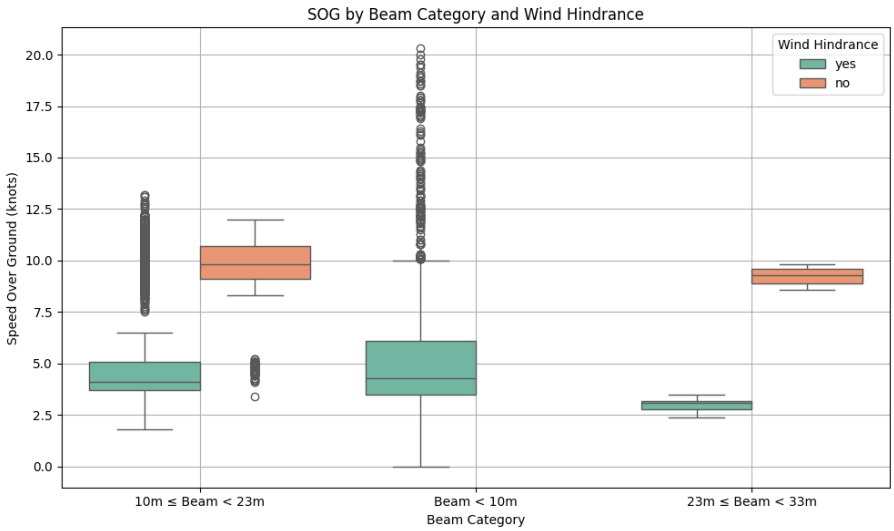


Figure 8: Rhein Haven: SOG by beam category under wind hindrance.

Summary: The results highlight that wind is the critical factor shaping vessel speed and maneuvering behaviour in the Rheinhafen:

- Inbound (upriver) traffic is especially slow and sensitive, suffering 15–20% reductions under headwinds.
- Outbound ships maintain higher speeds (∼10 kn) with negligible hindrance effects.
- Beam sensitivity is strong: small to mid-beam ships (10–33 m) experience the most pronounced slowdowns (to 3–4 kn).
- Turn probability is highest for inbound vessels under cross-winds (5.5%), while outbound vessels remain stable.

Comparison with Zhou's findings. Table 5 shows how these results align with Zhou et al. [24, 25].

Table 5: Comparison of Rhein Haven findings with Zhou et al. [24, 25].

Aspect	Our Findings	Zhou's Findings
Wind effect on speed	Large slowdown (4 to 6 kn)	Stronger impact in confined/shallow channels
Beam sensitivity	Very strong: 10–33 m beam vessels drop from 9–10 kn to 3–4 kn	Beamier mid-size ships more affected

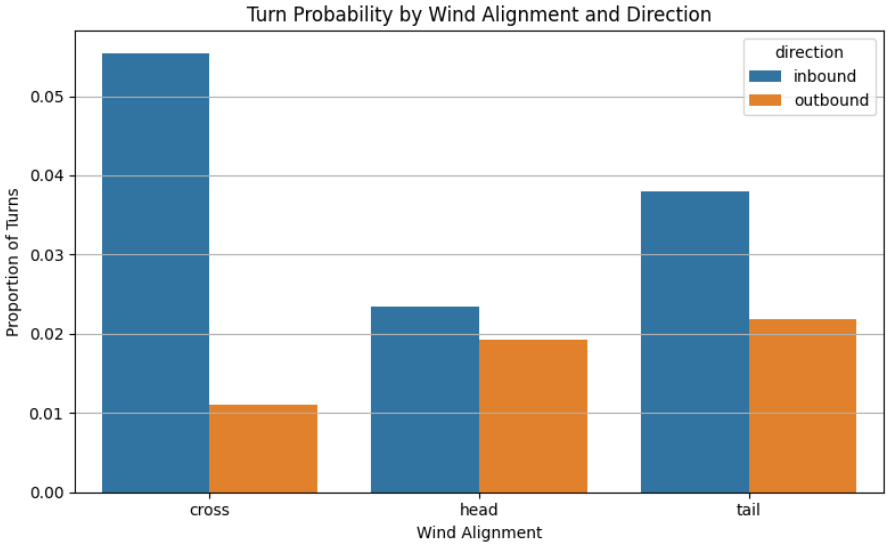


Figure 9: Rhein Haven: Turn probability by wind alignment (inbound vs outbound).

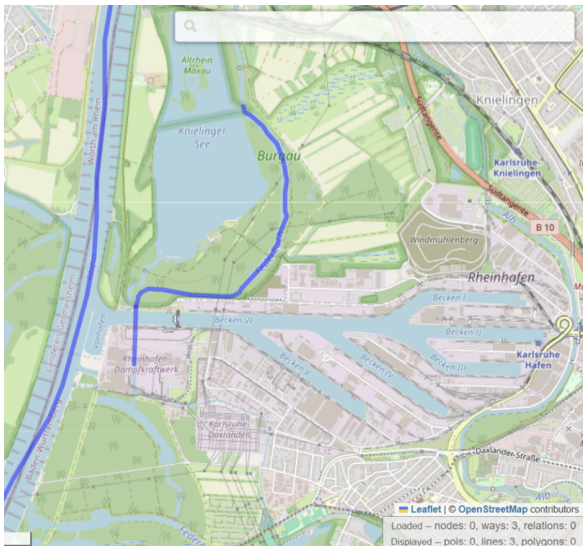
5.3.3 Relation to Literature

The findings align closely with prior work by Zhou et al. [24], who identified that **Current effects** dominate in open-water and tidal approaches (as confirmed in Cuxhaven). Whereas, **wind effects** dominate in inland rivers and confined ports (as confirmed in Rheinhafen). Lastly, the beam acts as a strong size proxy for sensitivity to hindrance, with narrower vessels experiencing more pronounced slowdowns.

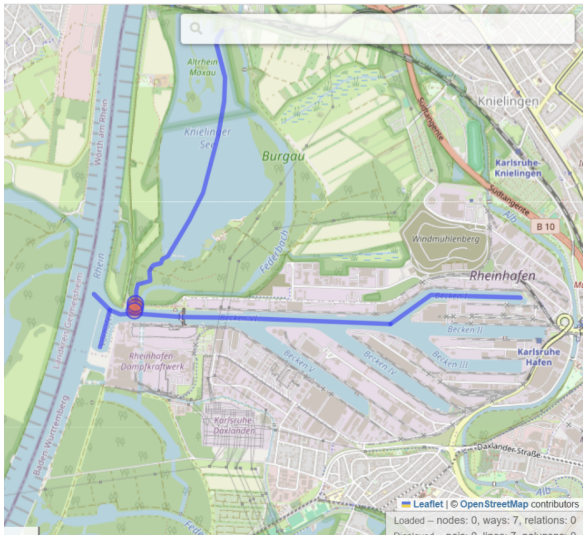
These parallels validate both the methodology and the empirical observations of Zhao et al. [25], while extending their framework in two important ways. First, whereas Zhao concentrated on a single straight fairway, our analysis compared two contrasting environments—the confined inland Rhine and the open tidal waters near Cuxhaven—demonstrating how geography shapes the manifestation of environmental hindrance. Second, we used cargo vessels in the 10- 23 m beam category as the basis for our MMG modeling, thereby directly linking the AIS-weather analysis to a physically grounded simulation for a representative ship class. In this way, our work contextualizes Zhao's framework in new environments and provides a vessel-specific application that strengthens its practical relevance.

5.4 Physics-Based Modeling for both regions

The physics-based model was used to simulate ship motion by directly applying hydrodynamic principles and Newton's laws of motion rather than relying on purely data-driven methods. The model incorporates propeller thrust, rudder lift and drag, as well as environmental forces from wind and currents. These forces are resolved into surge, sway, and yaw equations of motion, which are numerically integrated to predict the vessel's trajectory along the fairway. This approach ensures that the simulation reflects realistic ship dynamics and can be extended to different operating conditions.



(a) Rheinhafen fairway lines (River)



(b) Rheinhafen fairway lines (Canal)



(c) Cuxhaven fairway lines

Figure 10: Fairway lines used for constructing the line of sight (LOS). Top: Rheinhafen (river and canal). Bottom: Cuxhaven region.

5.4.1 Line of Sight

The line of sight (LOS) for the ship trajectory was generated from publicly available fairway data obtained using the Overpass Turbo platform (OpenStreetMap) and storing the coordinates as ordered latitude–longitude pairs. This centerline was then sorted to maintain a continuous sequence from the start to the end of the fairway. For each simulation run, the LOS served as the baseline route: the ship's position was projected onto the nearest segment of the centerline, and the cross-track error (CTE) was computed to quantify lateral deviation. This LOS construction

provided both a geometric guide for trajectory tracking and a reference against which control corrections could be applied. [9]

In the Rheinhafen area, separate fairway centerlines were extracted for both the river section and the canal network, since vessels may traverse either depending on operational requirements. For the Cuxhaven region, the fairway geometry covers a larger and more complex section of the Elbe River, including multiple branches and harbor approaches. By combining these datasets, the LOS was constructed to reflect realistic navigational routes in both inland and coastal waters. This ensured that the ship simulation could be applied to different environments—straight canal stretches as well as meandering river channels—allowing for a more comprehensive evaluation of the trajectory tracking and control algorithms.

5.4.2 Ship Parameters

To simulate ship motion realistically, a set of hydrodynamic and propulsion-related parameters was defined as detailed in Table 6. These parameters capture the influence of propeller thrust, rudder forces, and environmental effects such as wind and current. Each of them plays a role in the three-degrees-of-freedom (3-DOF) motion equations for surge, sway, and yaw. The values were either drawn from established naval architecture [10, 1] literature or approximated based on standard models [2, 5, 8] for inland and coastal vessels.

Importantly, some basic vessel information, such as length, beam, and draught, was directly extracted from the AIS (Automatic Identification System) data. However, AIS does not provide detailed hydrodynamic coefficients or propeller/rudder characteristics. These missing parameters were therefore estimated from literature values and standards and further refined through iterative simulation testing.

To further improve accuracy, the unknown parameters were refined through an Automatic System Identification and Global Optimization Process Based on the Covariance Matrix Adaptation Evolution Strategy (CMA-ES). CMA-ES is a stochastic, population-based optimization algorithm well-suited for nonlinear, multi-dimensional problems such as ship maneuvering parameter estimation. It iteratively adjusts the candidate parameter set by simulating trajectories and minimizing the error with respect to observed or reference data. A similar approach has been successfully demonstrated in recent work by Yasutake and Yoshimura (2023), where CMA-ES was applied to fine-tune the MMG (Maneuvering Mathematical Group) model parameters using a real-scale ship trajectory data.

Table 6: Ship parameters and their roles in the model.

Parameter	Symbol	Description / Role in Model
Hull drag coefficient	D	Represents quadratic resistance opposing surge motion.
Propeller diameter	D	Defines the size of the propeller, affecting thrust generation.
Propeller revolutions	n	Revolutions per second (or RPM) driving thrust.
Advance ratio	J	Ratio of inflow velocity to propeller rotation; used to compute K
Thrust coefficient	$K_T(J)$	Empirical relation defining propeller thrust for given J.
Thrust deduction factor	t_p	Accounts for hull resistance added due to propeller action.
Wake fraction	w	Reduces effective inflow velocity to the propeller due to hull wak
Rudder lift coefficient	C_L	Determines lateral force generated by rudder at given angle.
Rudder drag coefficient	C_D	Captures additional resistance due to rudder deflection.
Rudder area	S	Surface area of rudder affecting lift and drag forces.
Wind surge area	$A_{\text{wind, surge}}$	Projected area exposed to wind in surge direction.
Wind sway area	$A_{\text{wind ,sway}}$	Projected area exposed to wind in sway direction.
Wind drag coefficient	$C_{D,\text{wind}}$	Coefficient for wind resistance forces.
Current surge area	$A_{\text{current, surge}}$	Area exposed to water current in surge direction.
Current sway area	$A_{\text{current,sway}}$	Area exposed to water current in sway direction.
Current drag coefficient	$C_{D,\text{current}}$	Coefficient for current resistance forces.

These parameters were implemented in the simulation framework to compute propeller thrust, rudder-induced sway forces, hull resistance, and external disturbances. In cases where direct measurements or vessel-specific data were not available, values were adopted from standard references and refined through iterative testing in the simulation environment.

Parameter Fine-Tuning with CMA-ES

Several hydrodynamic coefficients were not directly available from AIS data or literature and therefore required estimation. To refine these unknown parameters, an Automatic System Identification approach was applied using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [23].

The optimization was run for 126 generations with a population size of 16, minimizing the trajectory error between simulated and AIS data. Figure 11 shows the convergence of the error over the generations. The error decreased from over 1200 m to approximately 594 m at generation 120, indicating that the tuned parameters significantly improved model accuracy.

The best-performing parameter vector at generation 120 was:

$$[9.73 \times 10^{-4},\ 3.43 \times 10^{3},\ 7.30 \times 10^{-2},\ 6.06 \times 10^{-2},\ 49.9,\ 9.77,\ 5.90 \times 10^{-3},\ 2.38,\ 30.1]$$

These correspond to the following model parameters:

- D – Propeller diameter

- K_T – Propeller thrust coefficient
- AR – Rudder aspect ratio
- C_L – Rudder lift coefficient
- $C_{D,\text{rudder}}$ – Rudder drag coefficient
- S – Rudder area
- C_D – Hull drag coefficient
- t_p – Thrust deduction factor
- w – Wake fraction

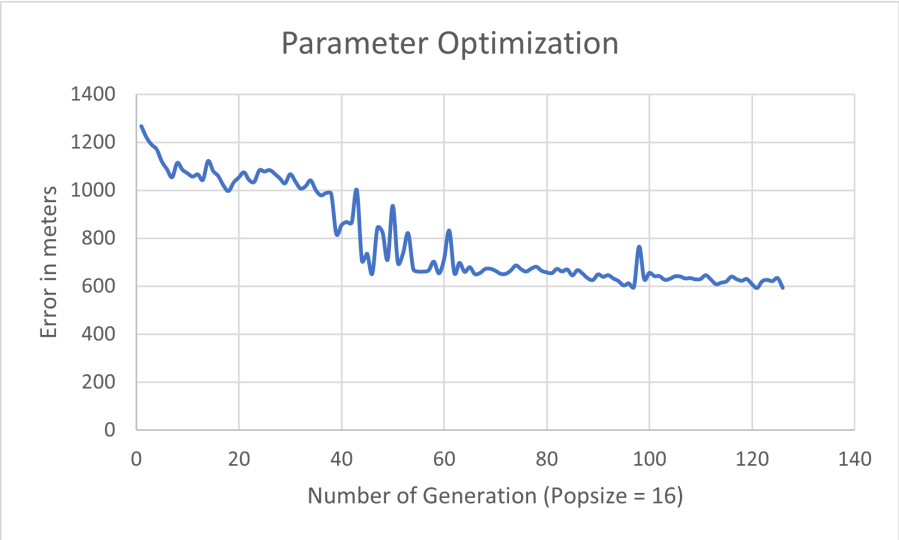


Figure 11: CMA-ES parameter optimization: error convergence over generations (population size = 16).

5.4.3 Model Creation

The ship model was developed as a physics-based 3-DOF framework (surge, sway, and yaw) by combining propeller, rudder, wind, and current forces. The approach is based on the MMG (Maneuvering Mathematical Group) model, which separates the contributions of each force component and is widely adopted in ship maneuvering studies.

1. Propeller Thrust

The propeller produces surge thrust as a function of advance ratio J and thrust coefficient $K_T(J)$:

$$T = \rho n^2 D^4 K_T(J), \quad J = \frac{u(1-w)}{nD}$$

where ρ is water density, n is propeller revolutions per second, D is propeller diameter, and w is the wake fraction.

The thrust coefficient $K_T(J)$ is typically obtained from experimental propeller series data, such as the Wageningen B-series, and is expressed as a polynomial function of the advance ratio J. For this study, an approximate quadratic relationship was used:

$$K_T(J) = aJ^2 + bJ + c$$

Where a, b, c are empirical coefficients determined from reference propeller curves or fine-tuned using optimization methods (CMA-ES). This formulation ensures that the propeller thrust decreases with increasing advance ratio, reflecting realistic propeller performance. [2]

2. Rudder Forces

The rudder generates sway force and yaw moment depending on rudder angle and flow conditions:

$$F_R = \tfrac{1}{2}\rho V^2 S(C_L \sin\alpha - C_D \cos\alpha), \quad N_R = F_R \cdot x_R$$

where S is rudder area, C_L and C_D are lift and drag coefficients, α is rudder angle of attack, and x_R is the rudder lever arm.[5]

3. Wind Forces

Wind loads were estimated using the Fujiwara model, which provides empirical force and moment coefficients for different ship types and wind angles:

$$F_{\text{wind}} = \tfrac{1}{2}\rho_{\text{air}} V_{\text{wind}}^2 A_{\text{proj}} C_{D,\text{wind}}$$

Wind velocity and direction were extracted from real-time weather data sources to ensure realistic simulation inputs.[6]

4. Current Forces

Current forces were modeled in a form similar to wind, using projected underwater areas and drag coefficients:

$$F_{\text{current}} = \tfrac{1}{2}\rho V_{\text{current}}^2 A_{\text{proj}} C_{D,\text{current}}$$

The current velocity and direction were obtained from oceanographic and weather databases in real time.[8]

5. Equations of Motion

The total forces in surge (X), sway (Y), and yaw (N) are combined as:

$$m(\dot{u} - vr) = X, \quad m(\dot{v} + ur) = Y, \quad I_z\dot{r} = N$$

Where u, v, r are surge velocity, sway velocity, and yaw rate, and I_z is yaw moment of inertia.[15]

6. Position Prediction

The ship's position and heading are updated through the kinematic relations:

$$\dot{x} = u\cos\psi - v\sin\psi, \quad \dot{y} = u\sin\psi + v\cos\psi, \quad \dot{\psi} = r$$

This allows continuous prediction of the vessel trajectory along the line of sight and evaluation of the cross-track error (CTE). [5]

5.4.4 Rudder Control System

The rudder control system was implemented to ensure that the vessel follows the predefined line of sight (LOS). The control law is based on two inputs: the heading error and the cross-track error (CTE), with a speed-dependent scaling applied to the CTE term. This allows the rudder to make effective corrections at low speeds while avoiding overreaction at higher speeds.

1. Heading Error
The heading error is the angular difference between the desired heading ψ_d (towards the next waypoint on the LOS) and the ship's actual heading ψ:

$$e_\psi = \psi_d - \psi$$

This term ensures that the ship gradually aligns its heading with the reference trajectory.[5]

2. Cross-Track Error (CTE)
The cross-track error is the shortest perpendicular distance between the ship's current position (x, y) and the LOS centerline:

$$e_y = (x - x_{\text{ref}})\, n_x + (y - y_{\text{ref}})\, n_y$$

where $(x_{\text{ref}}, y_{\text{ref}})$ is the closest point on the LOS and (n_x, n_y) is the normal vector of the LOS segment. This error reflects how far the ship has deviated laterally from the reference path.[5]

3. Control Law (Speed-Dependent)
The rudder angle δ was calculated as a linear combination of heading error and speed-adjusted cross-track error:

$$\delta = K_\psi e_\psi + K_y \frac{e_y}{u + \varepsilon}$$

where K_ψ is the heading error gain, K_y is the CTE gain, u is the ship's surge speed, and ε is a small constant to prevent division by zero.

At low speeds, dividing by u increases the weight of the CTE term, ensuring stronger course corrections. At high speeds, the CTE contribution is reduced, preventing excessive rudder activity and oscillations.[3]

4. Rudder Limits
To account for physical constraints, the rudder deflection was bounded by a maximum angle:

$$|\delta| \leq \delta_{\max}$$

This limit prevented unrealistic or unstable steering inputs.[14]

5. Integration with Simulation
At each simulation time step, the rudder angle was computed from the above control law. The resulting rudder force and yaw moment were then applied in the MMG equations of motion, enabling the vessel to reduce both heading error and cross-track error while following the fairway centerline.

5.5 Evaluation

5.5.1 Objective and Approach

The project set out to evaluate the performance of a vessel trajectory simulation model under prescribed conditions. Specifically, the aim was to measure how accurately the model could replicate known historic trajectories when provided with correct boundary conditions (start and end coordinates) and environmental forcing data (wind, and for open water, current), taken from track segments with locally consistent conditions varying only within 3–5 units, rather than averaged across the entire track.

The methodology combined:

1. Coordinate transformation of historic and simulated tracks into local Cartesian coordinates for precise geometric comparison.
2. Cross-Track Error (XTE) analysis to quantify positional deviations.
3. Path Similarity Index (PSI) analysis to evaluate the realism of vessel dynamics and turning behavior.

Figures 12–17 show the transformed trajectories and corresponding cross-track error distributions for all six test tracks.

5.5.2 Results

Cross-Track Error (XTE) Table 7 summarizes the statistical error metrics across the six tracks.

Table 7: Statistical Cross-Track Error (XTE) results across all tracks.

Location	Track	Mean XTE (m)	95th %ile (m)	Max (m)	Std. Dev. (m)
Cuxhaven	H3	10.7	38.9	55.2	11.4
Rhein	H2	12.1	20.3	29.6	6.7
Cuxhaven	H2	28.5	73.6	102.4	21.9
Cuxhaven	H1	31.9	82.3	115.8	24.3
Rhein	H3	19.7	49.5	68.1	15.6
Rhein	H1	34.8	82.3	121.2	25.1

Path Similarity Index (PSI) Table 8 shows the Path Similarity results.

5.5.3 Interpretation

1. **Validated physics engine:** Excellent performance in Cuxhaven H3 and Rhein H2 validates the simulation's hydrodynamic core.

Table 8: Path Similarity results across all tracks.

Location	Track	Similarity Index	Category
Cuxhaven	H3	0.928	Near-perfect
Rhein	H2	0.958	Near-perfect
Cuxhaven	H2	0.673	Moderate
Cuxhaven	H1	0.556	Moderate
Rhein	H3	0.862	High
Rhein	H1	0.556	Moderate

2. **Controller fidelity:** In confined waters or when currents are negligible, the controller achieves sub-20 m errors, comparable with professional pilotage standards.

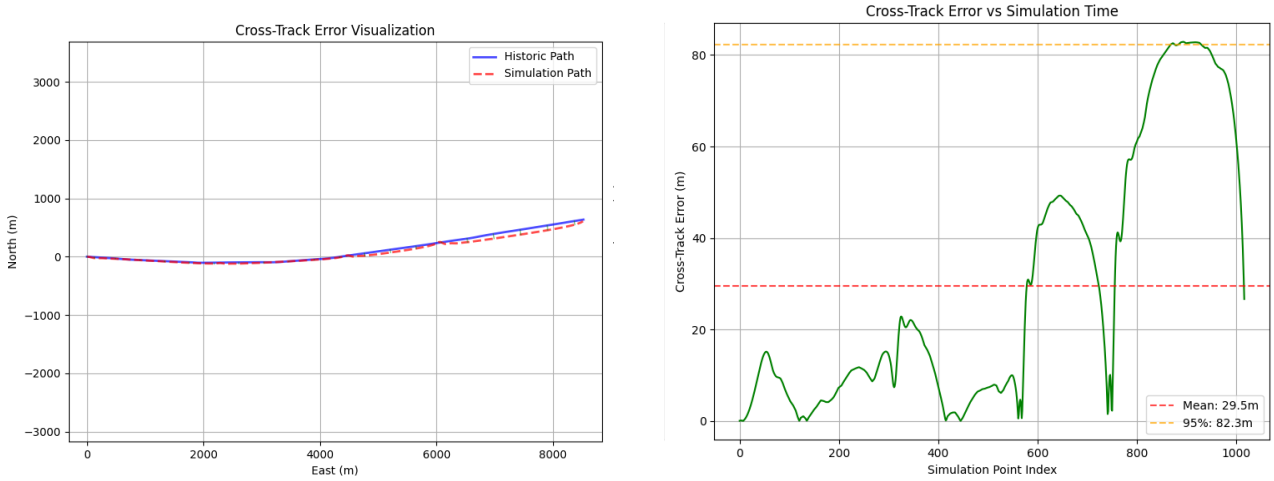
3. **Input sensitivity:** The largest discrepancies occurred where the selected environmental segments did not fully capture local variability — for example, unresolved tidal currents in Cuxhaven or transient gusts in Rhein.

4. **Worst-case matters most:** Median errors are always small (4–25 m). The differentiator is the 95th percentile XTE, which captures rare but significant deviations.

5. **Behavioral realism:** Path Similarity analysis reveals when the model "looks right." High error tracks often fail not just in position but in replicating the decision-making style of human navigators.

6. **Turn-handling limitations:** The Rhein H1-S1 track illustrates the model's weakest scenario: a sharp port turn resulted in nearly 90 m XTE, showing that the controller underestimates aggressive helm corrections in confined waters.

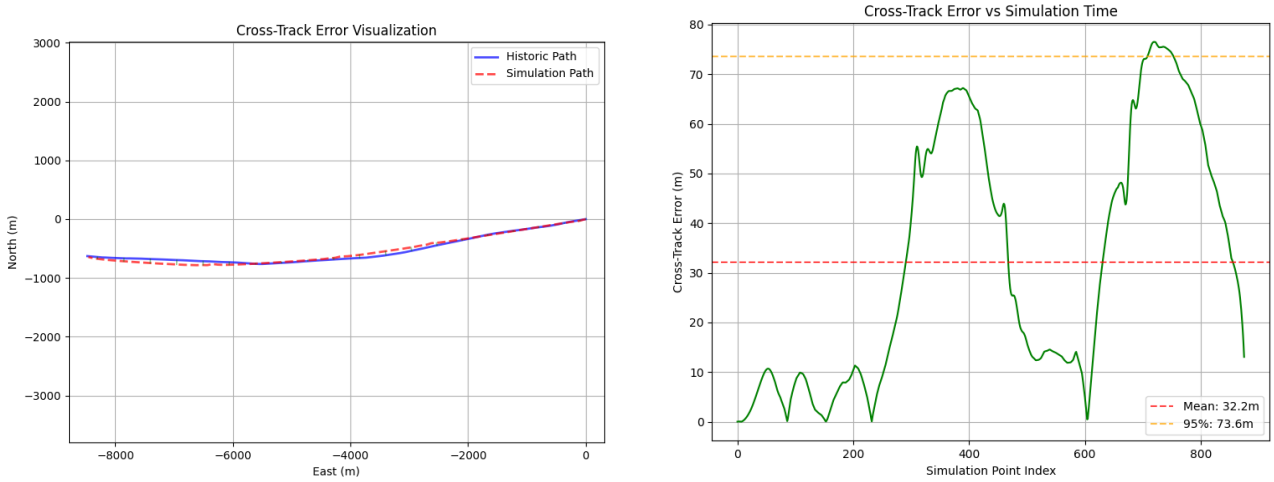
5.5.4 Visual Results



(a) Historic and simulated trajectory

(b) Cross-Track Error graph

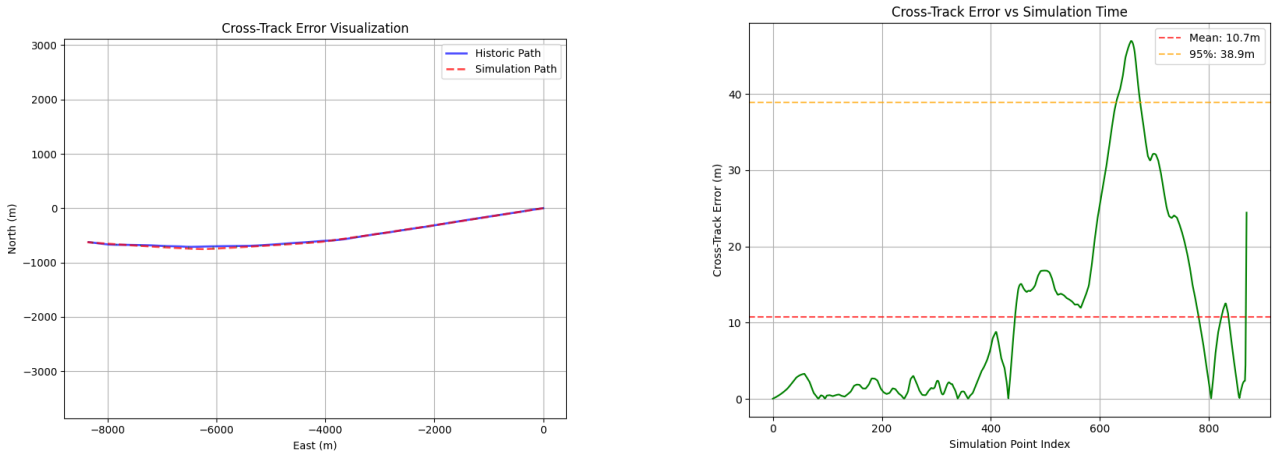
Figure 12: Cuxhaven H1: trajectory and Cross-Track Error (CTE). Moderate similarity (0.556) with large deviations up to 115 m in tail end.



(a) Historic and simulated trajectory

(b) Cross-Track Error graph

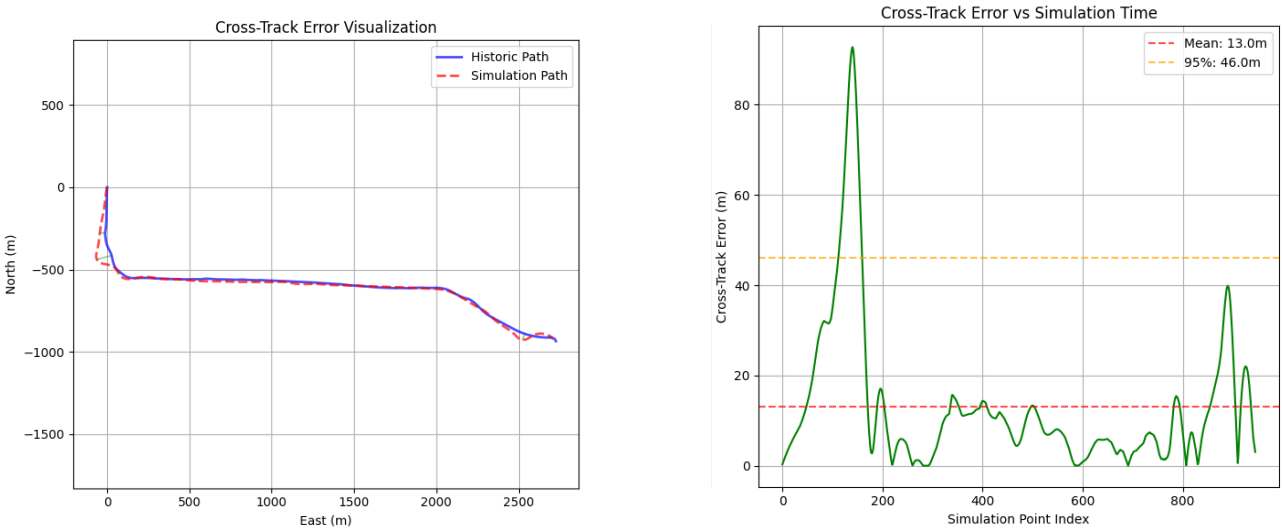
Figure 13: Cuxhaven H2: trajectory and CTE. Large deviations (95th %ile = 73 m) suggest unresolved current variability in the estuary contributed to higher errors.



(a) Historic and simulated trajectory

(b) Cross-Track Error graph

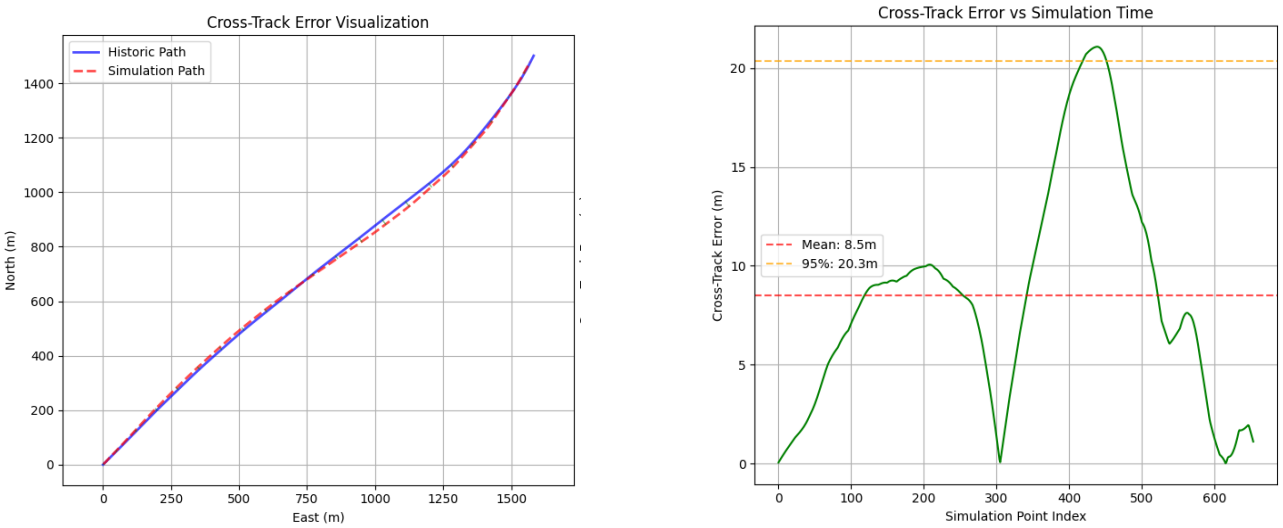
Figure 14: Cuxhaven H3: trajectory and CTE. Near-perfect match (Similarity = 0.928).



(a) Historic and simulated trajectory

(b) Cross-Track Error graph

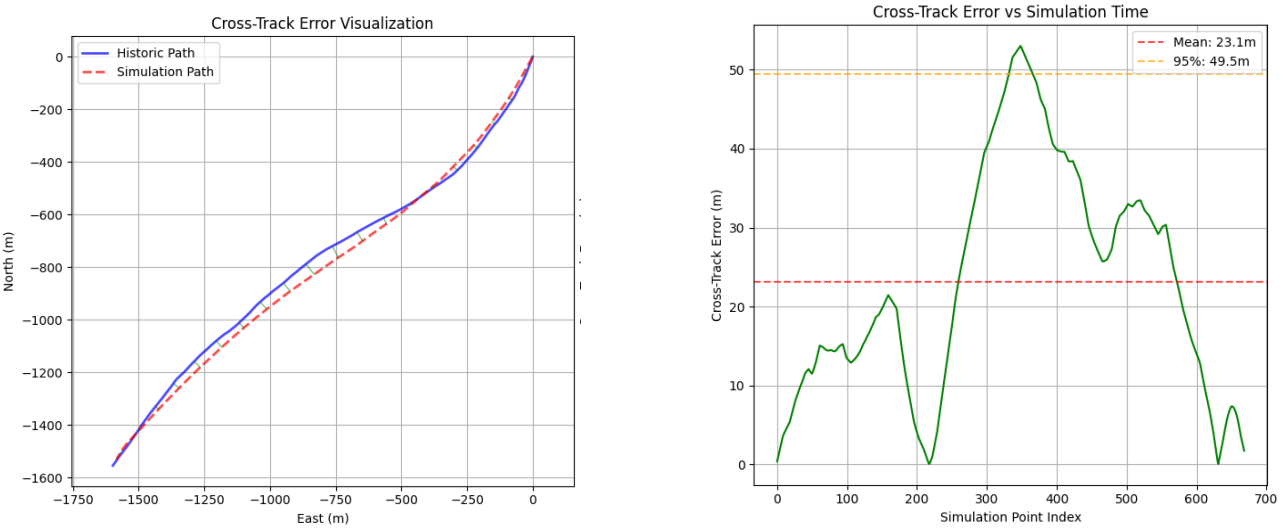
Figure 15: Rhein H1-S1: trajectory and CTE. A sharp port-side turn caused nearly 90 m deviation, highlighting the controller's difficulty in handling confined-water turns.



(a) Historic and simulated trajectory

(b) Cross-Track Error graph

Figure 16: Rhein H2: trajectory and XTE. Excellent controller performance with sub-20 m error and near-perfect similarity (0.958).



(a) Historic and simulated trajectory

(b) Cross-Track Error graph

Figure 17: Rhein H3: trajectory and XTE. High similarity (0.862) with lateral drift due to local wind effects.

5.6 Front End UI Development

A dedicated front-end interface was developed to support visualization and interaction with the physics-based simulation model for both regions (Rheinhafen and Cuxhaven). The interface first allows the user to select the desired region. Once a region is chosen, a **folium-based**

interactive map is displayed, constrained by the predefined bounding box of that region. Within this map, users can specify the trajectory by selecting start and end points.

Upon selection, the interface invokes the physics-based model and renders the simulated trajectory using the **Plotly visualization library**. The user is provided with several customization and interaction options, including adjustment of the simulation playback speed, export/download of the generated trajectory data in CSV format, and the ability to reset the selected points in order to simulate multiple trajectories with different start and end points.

This front-end implementation ensures reproducibility and facilitates exploration of the modeled system by providing an intuitive, interactive, and data-driven visualization environment. The front-end UI is also scalable and versatile. In future, if the team intends to develop other forms of models or even data-driven models using Machine Learning or Deep learning approaches, the UI is flexible enough to handle these changes.

5.7 Team Contribution and Collaboration

This project was carried out through collaboration among all three team members — Aishwarya, Nishanth, and Vineet — with a clear division of responsibilities and shared ownership of critical phases.

- **All three Members:** Literature Review, EDA and documenting their respective parts in the project.
- **Aishwarya:** Setting up Github repository, Data cleaning and analysis of Environmental factors.
- **Nishanth:** Physics-based Modeling and refinements.
- **Vineet:** Developing UI with further refinements and setting up Overleaf environment for the Scientic Report and proofreading.
- **Collaboration Between Tasks:** Aishwarya and Nishanth worked jointly on testing and evaluation of trajectory performance, ensuring consistency between empirical AIS-weather results and physics-based model outputs. Vineet and Nishanth collaborated to test the physics based model for edge cases for trajectory refinements.

6 Comparison of Project Plan and Execution

The execution of the project largely adhered to the initial timeline and work packages established in the project plan. The majority of the phases and milestones were achieved on schedule, and the project's primary scientific and technical objectives were successfully met. The team's workflow and collaboration remained effective throughout the project duration, allowing for consistent progress.

A minor deviation from the original schedule occurred during the core modeling phase. The development of the physics-based model and the subsequent validation and refinement of the simulated trajectories proved to be more complex than anticipated. This critical phase extended through the end of August, slightly beyond its planned completion date. This extension was necessary to ensure the robustness and accuracy of the model, which formed the foundation of the entire project.

As a direct consequence of this shift, the final development stage, which included the UI development and its refinement, was pushed into the first two weeks of September. Despite this delay in the preceding phase, the UI was completed efficiently. Overall, this timeline adjustment was managed without significant disruption to the project's overall momentum or its final deliverables.

6.1 Work that did not lead to the intended results

All core work packages produced their intended results. While the model refinement process involved iterative testing where some parameter sets were discarded, this is a standard part of scientific model development and not considered a deviation from the intended outcome. Every phase, from data analysis to the final UI, contributed successfully to the project's goals.

6.2 Necessary replanning done during project duration

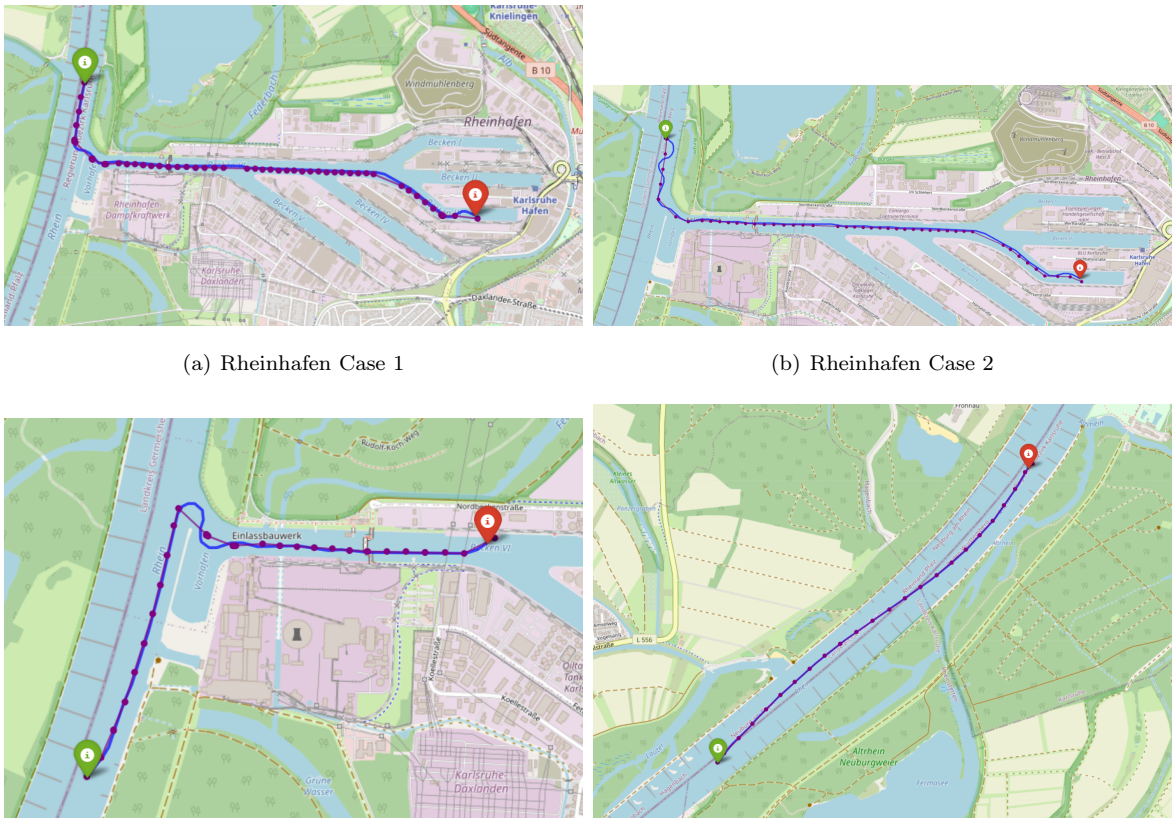
No major replanning was required during the project. The minor schedule adjustment for the UI development was a sequential shift rather than a fundamental change in the project's structure or objectives. The team coped with the challenge of the extended modeling phase by maintaining clear communication and ensuring that the person incharge of UI development was prepared to begin work as soon as the finalized model was available. This proactive management ensured that the project concluded successfully without the need for a formal replanning process. In reflection, the project journey was a success, achieving its aims through consistent effort and effective adaptation to minor scheduling challenges.

7 Usability of the Results

Trajectory Visualization Results

To qualitatively evaluate the performance of the physics-based model, interactive trajectory visualizations were generated using the `Folium` library. These maps show how the simulated vessel follows the fairway centerline in both Rheinhafen and Cuxhaven study regions. Start and end markers are indicated in green and red, respectively, while the simulated trajectories are drawn as polylines along the line-of-sight (LOS) centerline.

Rheinhafen: Figures 18(a)–18(d) illustrate simulated trajectories within the inland port region of Rheinhafen. These cases demonstrate how the ship is guided along both straight and curved canal sections. The trajectories closely follow the LOS centerline, and deviations are corrected using the rudder control law, producing smooth and realistic paths that remain confined within the fairway boundaries.



(a) Rheinhafen Case 1

(b) Rheinhafen Case 2

(c) Rheinhafen Case 3

(d) Rheinhafen Case 4

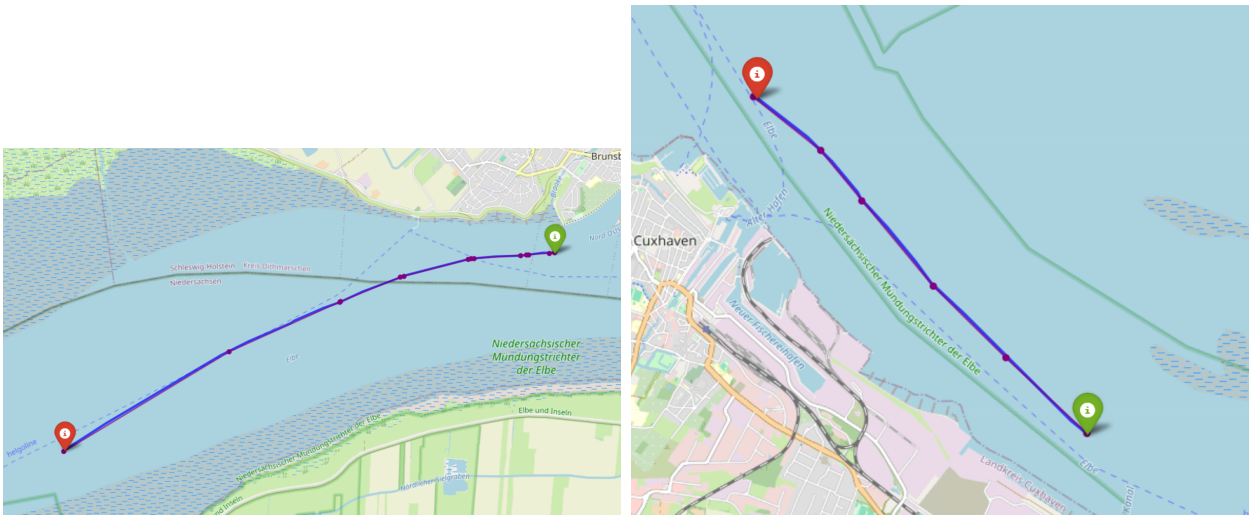
Figure 18: Simulated trajectories in Rheinhafen generated using Folium.

Cuxhaven: Figures 19(a)–19(b) show trajectory visualizations in the Cuxhaven region. Unlike Rheinhafen, this estuarine environment involves wider fairways and tidal influences. The simulated trajectories demonstrate stable guidance along the LOS, highlighting the ability of the model to reproduce realistic vessel paths even in open water conditions.

These visualizations confirm that the physics-based model, coupled with LOS guidance and rudder control, can replicate realistic trajectories across different waterway types. In confined inland canals (Rheinhafen) as well as open estuary conditions (Cuxhaven), the vessel remains aligned with the fairway centerline and deviations are effectively corrected.

7.1 Practical Applications of the Results

The outcomes of this study have two main areas of practical application:



(a) Cuxhaven Case 1 (b) Cuxhaven Case 2

Figure 19: Simulated trajectories in Cuxhaven generated using Folium.

Simulation: The physics-based trajectory model can be applied in:

- **Autonomous navigation research**, where realistic ship dynamics under wind and current need to be tested and validated.
- **Traffic management and port planning**, by simulating vessel speed loss and drift in confined or open waters.
- **Safety analysis**, through the quantification of cross–track error and deviation under hindrance scenarios.
- **Training and decision support**, where realistic trajectories under given environmental inputs can be reproduced for operators or pilots.

Data Analysis: The AIS–weather coupling pipeline has value for:

- **Benchmarking ship behavior**, by establishing empirical speed reductions (ΔSOG) under hindered vs. unhindered conditions.
- **Operational planning**, such as improved ETA predictions that account for weather effects.
- **Risk and insurance**, where quantified drift and deviation statistics can be used in liability or safety assessments.

Stakeholders that benefit from these applications include pilots, operators, port and VTS authorities, logistics planners, insurers, and researchers in vessel dynamics and autonomy.

7.2 Constraints and Limitations

Despite these applications, several limitations constrain the scope of the present work:

- **Data Horizon:** The analysis was based on only 15 days of AIS and meteorological data, which limits representativeness across different seasons, tidal cycles, and extreme events.
- **Scalability of Methodology:** Polygon-based area selection was used to define study regions (Rhein Haven and Cuxhaven). While effective for isolating unhindered waterway segments, this introduces inductive bias and constrains scalability to more complex or irregular port geometries.
- **Environmental Forcing Coverage:** The study incorporated wind, current, and visibility, but did not include sudden wind gusts, tidal shear layers, or localized eddies. Such fine-scale effects can significantly alter vessel motion and were outside the present scope. Coarse weather/ocean data may not capture local variations.
- **Encounter Filtering:** Vessel encounters were deliberately excluded to isolate the influence of environmental factors. While this clarified weather impacts, it omits crossing, overtaking, and head-on situations that dominate behavior in congested waterways.
- **Model Assumptions:** The MMG-based physics model was implemented with fixed representative ship parameters (mass, inertia, drag coefficients). This simplifies computation but reduces generalizability across different vessel classes and loading conditions. The effects of weather on different types of ships, although analyzed in AIS data, were not dynamically integrated into the simulation.

8 Conclusions

This project successfully investigated the influence of environmental factors on vessel trajectories by integrating empirical analysis of AIS data with a physics-based simulation model. By applying established methodologies to two geographically distinct waterways—the confined inland Rhine and the open tidal waters of Cuxhaven—we sought to understand how local context mediates the impact of wind and current on ship navigation. The key achievement of this work is the development of a hybrid modeling pipeline that quantifies environmental hindrance and simulates realistic vessel motion. Our results demonstrate that while the fundamental effects of wind and current are consistent, their magnitude and nature are significantly shaped by waterway geometry. We also affirmed that a physics-based approach provides a robust alternative when extensive, high-quality datasets for training complex machine learning models are unavailable.

However, the study has several limitations that suggest clear directions for future work. The analysis was constrained by a 15-day data horizon, which does not capture full seasonal or tidal variabilities. To improve, future research should incorporate longer-term datasets and integrate

more complex phenomena such as vessel encounters, which were deliberately excluded to isolate environmental effects. Furthermore, data-driven appraoches can also be used for predicting and simulating shipment trajectories based on prior patterns.

In conclusion, this project successfully demonstrates a methodology for analyzing and simulating weather-driven vessel trajectories in varied maritime environments. It underscores the critical interplay between environmental forces and geography, paving the way for more accurate, reliable, and context-aware predictive models essential for the future of maritime safety and autonomous navigation.

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